

# Vikram Govindarajan

Kalpakkam, Tamil Nadu, India

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## SUMMARY

Nuclear engineer with over 10 years of experience in thermal-hydraulics, reactor safety analysis, and scientific computing for sodium-cooled fast reactors (SFRs). Proven track record of developing in-house simulation frameworks for plant dynamics. Passionate about bridging classical reactor simulations with modern techniques including high-performance computing, machine learning, and simulation process & data management.

## EDUCATION

### Homi Bhabha National Institute (HBNI), Mumbai

*Ph.D. in Engineering Sciences*

Kalpakkam, TN

*Jan. 2024 – Dec. 2027 (expected)*

### Homi Bhabha National Institute (HBNI), Mumbai

*Master of Technology in Mechanical Engineering, Specialization in Fast Breeder Reactor and Recycle Science & Technology*

Kalpakkam, TN

*Aug. 2014 – Dec. 2016*

### College of Engineering, Guindy (CEG)

*Bachelor of Engineering in Mechanical Engineering*

Chennai, TN

*Jul. 2010 – Apr. 2014*

## EXPERIENCE

### Scientific Officer

*Indira Gandhi Centre for Atomic Research (IGCAR), DAE, Govt. of India*

Aug. 2014 – Present

*Kalpakkam, TN*

- Performed plant dynamics and safety analysis for indigenous Fast Breeder Reactors (FBRs)
- Conducted system thermal-hydraulic analysis for sodium-cooled reactor systems
- Contributed to the development of Severe Accident Management Guidelines (SAMG) for fast reactors
- Participated in IAEA Coordinated Research Projects (CRPs) related to reactor analysis: FFTF and S-Allegro benchmarks
- Mentored undergraduate students and trainees in reactor simulations, and scientific computing

## PROGRAMMING/SOFTWARE SKILLS

**Programming Languages:** Python, C/C++, Fortran, MATLAB

**Engineering Software:** Flownex

**Developer Tools:** Git, Sphinx, L<sup>A</sup>T<sub>E</sub>X, etc.

**Scientific Libraries:** NumPy, Scipy, Matplotlib, CoolProp, etc.

## PROJECTS

### PINET: Thermal-Fluid Simulation Framework

- Developing an in-house general-purpose plant dynamics code for reactor system analysis
- Implemented parallel solvers and domain decomposition techniques using OpenMP, MPI
- Exploring machine learning approaches (PINNs) for transient prediction

## PUBLICATIONS

Journal articles: **9** (7 published, 2 in press)    Conference papers: **10+**

*Selected Journal Articles:*

- G. Vikram, K. Natesan, M. Rajendrakumar, S.P. Pathak, “Development and validation of transient thermal-hydraulic model for SFR steam generators,” *Annals of Nuclear Energy*, vol. 180, p. 109490, 2023.
- G. Vikram, A. Riyas, A. Bachchan, M. Rajendrakumar, K. Devan, K. Natesan, “Modeling of sodium-cooled fast reactor core – A sensitivity study on number of core channels and axial meshes,” *Nuclear Engineering and Design*, vol. 403, p. 112136, 2023.

## LANGUAGES

**Tamil:** Native

**English:** Professional